The descent set of the tableau of a matching is its set of short chords

Avichai Marmor

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Editorial note. The counted statement is the author's labelled proposition environment at publisher-source.tex lines 897-900; its proof environment opens at line 901, immediately after \end{proposition}, so the pairing is by adjacency and the retained proof range 901-906 is the whole environment.

Editorial note. The lemma lem:properties of core, with its three numbered items (source lines 873-881), is retained as context because the counted proof cites its first item at source line 905; it is the only outside label of the retained proof.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label prop:F statistic preserving.

Editorial note. r002 note: r001 failed with an undefined reference to def:restriction of matching, which the retained lem:properties of core context cites at source line 879; r002 adds exactly that definition environment as a further context and changes nothing else.

Editorial scope. One original result of the pinned publisher article, counted once: the descent set of the tableau of a matching is its set of short chords (author label prop:F statistic preserving).

Allowed standard prerequisites (editorial).

- Core and stable vertices. The core of a matching is the matching induced on its stable vertices and the two-row tableau T(m) records the unstable chords in its second row.. They define the tableau whose descent set is computed.
- Short chord and descent set. A chord (i,i+1) is short when no vertex lies strictly between i and i+1, and the descent set of a tableau is the set of indices i with i+1 in the row below i.. They are the two sets that the proposition identifies.
- Stability of a chord intersecting a non-crossing chord. A chord that intersects another chord is stable.. It is the contradiction used in the proof of the reverse inclusion.

Definition 0.1. Given a matching $m \in \mathcal{M}_N$, we say that a set $S \subseteq [N]$ is m-invariant if for every chord $(i,j) \in m$ we have $i \in S$ if and only if $j \in S$. Moreover, for a matching m and an m-invariant set S we define the restriction of m to S, denoted $\mathrm{res}_S(m)$, to be a matching on S which is obtained by removing all the vertices not in S from m.

Lemma 0.2. The reduction process has the following properties:

(1) If a chord intersects another chord then it is stable.

(2) A chord (i, j) is stable if and only if there exists a stable vertex in $\{i+1, \ldots, j-1\}$.

(3) Given a matching m and $i < j$, if the set $[i, j]$ is m-invariant (as defined in Definition 0.1) and the restricted matching $\mathrm{res}_{[i,j]}(m)$ is perfect and non-crossing, then $\ell \notin \mathrm{Stable}(m)$ for all $\ell \in [i, j]$.

Proposition 0.3. Let $m \in \mathcal{M}_{N,f}$ be a matching, and denote $T := T(m)$. Then $\mathrm{Des}\, T = \mathrm{Short}\, m$.

Proof. Let $i \in \mathrm{Short}\, m$ be an index representing a short chord $(i, i+1) \in m$. The chord $(i, i+1)$ is unstable, so we may deduce from the definition of $T(m)$ that $i \in \mathrm{row}_1(T)$ and $i+1 \in \mathrm{row}_2(T)$. Therefore, $i \in \mathrm{Des}\, T$.

On the other hand, let $i \in \mathrm{Des}\, T$ be a descent of T. This implies that $i \in \mathrm{row}_1(T)$ and $i+1 \in \mathrm{row}_2(T)$. Since $i+1 \in \mathrm{row}_2(T)$, we obtain that there exists $j < i+1$ such that $(j, i+1) \in m$ is an unstable chord. Since the chord $(j, i+1)$ is unstable and $j \leq i \leq i+1$, we obtain by Lemma 0.2 (part 2) that i is an endpoint of an unstable chord of m. Since $i \in \mathrm{row}_1(T)$, we obtain that i opens an unstable chord (i, j') of m for some $j' > i$. If $(i, i+1) \notin m$ then $j < i$ and $j' > i+1$, and we obtain that the unstable chord $(j, i+1)$ intersects the chord (i, j'), contradicting Lemma 0.2 (part 1). Therefore, we may conclude that $(i, i+1) \in m$ and $i \in \mathrm{Short}\, m$. □